brms_reg_school_nesting Diagnostics

Ibrahim Osumanu

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```r
brms_reg_model_production <- readRDS("~/OneDrive/Bayesian
Inference/brms_reg_model_production.rds")
summary(brms_reg_model_production)
```

```
##  Family: bernoulli
##   Links: mu = logit
## Formula: is_sped ~ reg + (1 | Schl)
##    Data: df_reg_clean (Number of observations: 139601)
##   Draws: 4 chains, each with iter = 5000; warmup = 2500; thin = 1;
##          total post-warmup draws = 10000
##
## Multilevel Hyperparameters:
## ~Schl (Number of levels: 222)
##               Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sd(Intercept)     0.72      0.04     0.65     0.81 1.01      865     1861
##
## Regression Coefficients:
##                             Estimate Est.Error l-95% CI u-95% CI Rhat
Bulk_ESS
## Intercept                      -1.87      0.05    -1.96    -1.77 1.01
385
## regAfricanAmerican_Female       0.16      0.03     0.10     0.21 1.00
4177
## regAfricanAmerican_Male         0.78      0.03     0.73     0.83 1.00
3775
## regAmericanIndian_Female        0.01      0.18    -0.36     0.36 1.00
11559
## regAmericanIndian_Male          0.65      0.15     0.35     0.94 1.00
12132
## regAsian_Female                -1.21      0.10    -1.41    -1.03 1.00
11211
## regAsian_Male                  -0.25      0.06    -0.37    -0.12 1.00
9054
## regHawaiian_Female              0.12      0.31    -0.53     0.69 1.00
11967
## regHawaiian_Male                0.14      0.31    -0.49     0.70 1.00
13402
## regHispanicDLatino_Female      -0.14      0.03    -0.21    -0.08 1.00
5521
## regHispanicDLatino_Male         0.44      0.03     0.38     0.50 1.00
4968
## regMultiMRacial_Female         -0.05      0.05    -0.16     0.05 1.00
```

```
8515
## regMultiMRacial_Male          0.64      0.04     0.55     0.73 1.00
7660
## regWhite_Male                 0.56      0.02     0.51     0.61 1.00
4033
##                           Tail_ESS
## Intercept                      812
## regAfricanAmerican_Female     6441
## regAfricanAmerican_Male       6527
## regAmericanIndian_Female      7493
## regAmericanIndian_Male        7769
## regAsian_Female               7591
## regAsian_Male                 7346
## regHawaiian_Female            6807
## regHawaiian_Male              7018
## regHispanicDLatino_Female     6813
## regHispanicDLatino_Male       6961
## regMultiMRacial_Female        7345
## regMultiMRacial_Male          7148
## regWhite_Male                 5954
##
## Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

```
library(performance)
```

```
## Warning: package 'performance' was built under R version 4.4.3
```

```
icc_results<-icc(brms_reg_model_production)
print(icc_results)
```

```
## # Intraclass Correlation Coefficient
##
##     Adjusted ICC: 0.137
##   Unadjusted ICC: 0.132
```

```
library(bayesplot)
```

```
## Warning: package 'bayesplot' was built under R version 4.4.3
```

```
## This is bayesplot version 1.11.1
```

```
## - Online documentation and vignettes at mc-stan.org/bayesplot
```

```
## - bayesplot theme set to bayesplot::theme_default()
```

```
##    * Does _not_ affect other ggplot2 plots
```

```
##    * See ?bayesplot_theme_set for details on theme setting
```

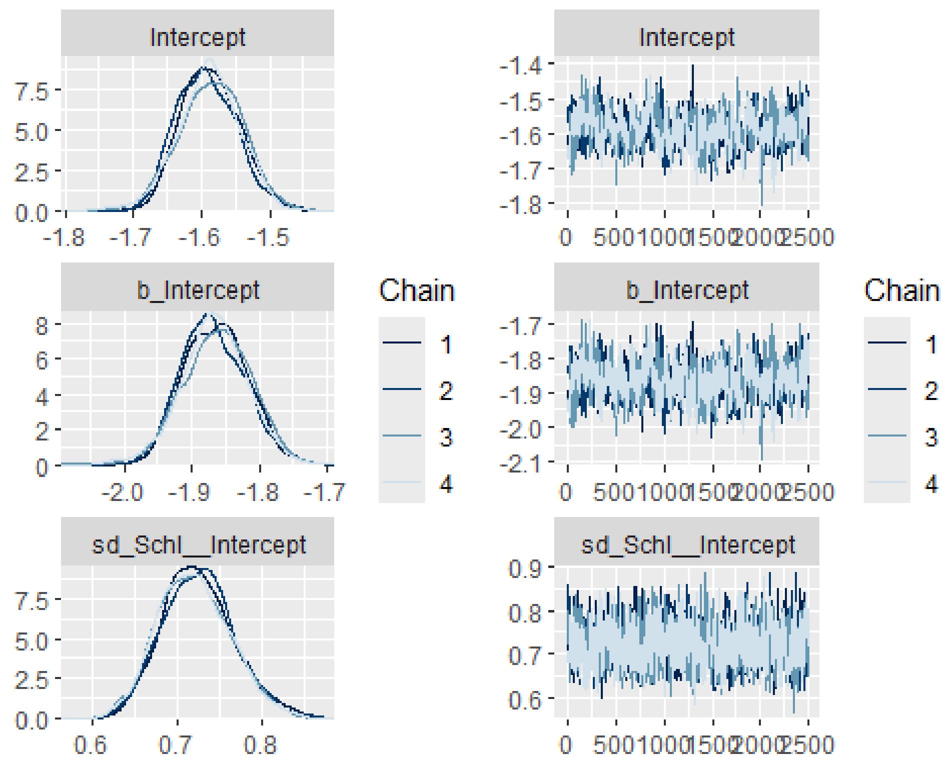
```
library(ggplot2)
```

```
## Warning: package 'ggplot2' was built under R version 4.4.3
```

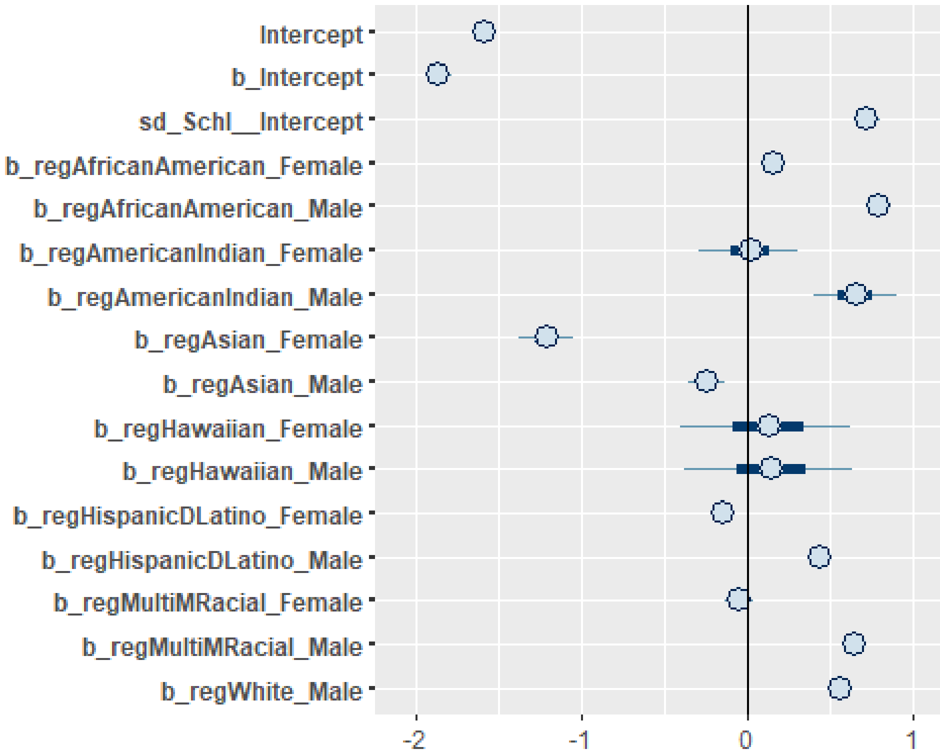
```
# Select your main parameters (Global Intercept and Variances)
# 'sd_Group__Intercept' is the typical brms name for random intercept SD
params <- c("Intercept","b_Intercept", "sd_Schl__Intercept")

params_1<-c("Intercept","b_Intercept",
"sd_Schl__Intercept","b_regAfricanAmerican_Female",
"b_regAfricanAmerican_Male",
"b_regAmericanIndian_Female","b_regAmericanIndian_Male","b_regAsian_Female","
b_regAsian_Male","b_regHawaiian_Female","b_regHawaiian_Male","b_regHispanicDL
atino_Female","b_regHispanicDLatino_Male","b_regMultiMRacial_Female","b_regMu
ltiMRacial_Male","b_regWhite_Male")

# Generate the side-by-side plot
mcmc_combo(brms_reg_model_production,
                pars = params,
                combo = c("dens_overlay", "trace"))
```



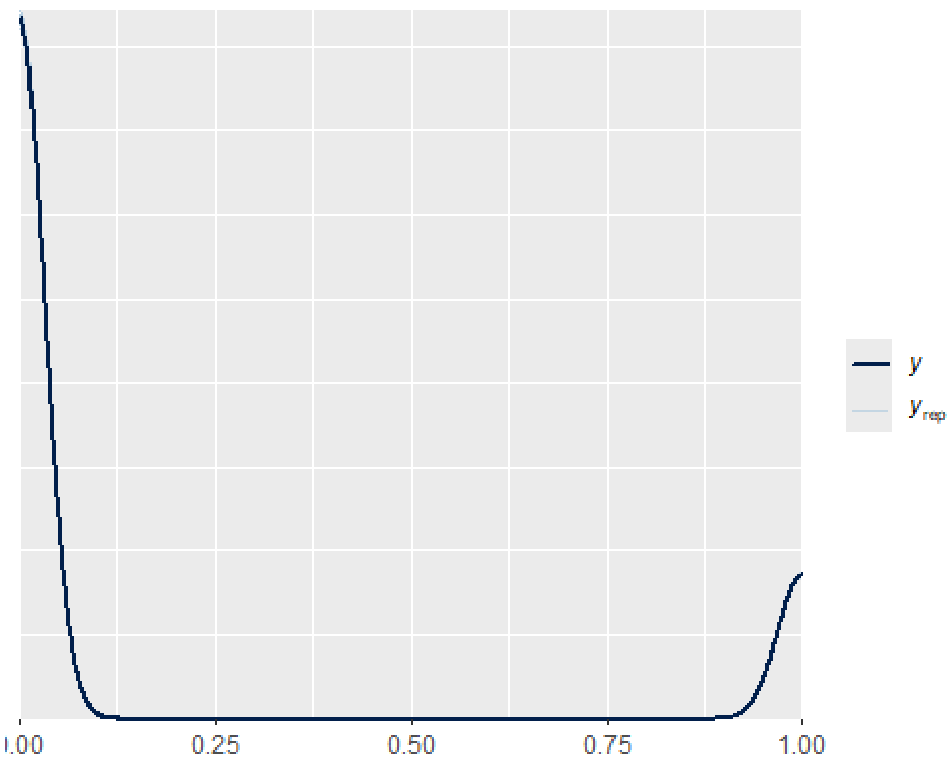

```
# This shows the 50% (thick line) and 95% (thin line) credible intervals
mcmc_intervals(brms_reg_model_production,
                              pars = params_1) +
               vline_0() # Adds a vertical line at zero for significance
testing
```



Posterior Predictive Checks

```
library(bayesplot)
pp_check(brms_reg_model_production, nsamples = 50)

## Warning: Argument 'nsamples' is deprecated. Please use argument 'ndraws'
## instead.
```


.00
0.25
0.50
0.75
1.00
y
y rep